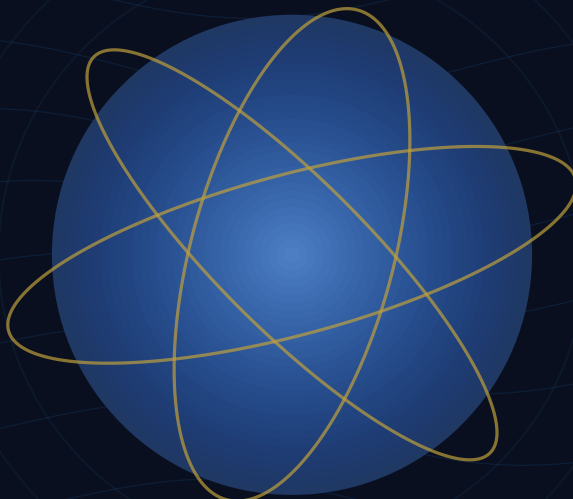


VOLUME I

DYNAMIC REALITY FIELD THEORY

Vol. I — Foundation, Geometry and Initial Predictions

$$\frac{dR}{dt} = c_R^2 * \nabla^2 R - \lambda_R * R + S_R(x,t)$$



ADVAIT PANDYA

Theoretical Physics Manuscript • First Edition • 2026

Dynamic Reality Field Theory (DRFT)

Vol. I — Foundation, Geometry and Initial Predictions

Advait Pandya

Independent Theoretical Physics Research • 2026

ABSTRACT

We introduce Dynamic Reality Field Theory (DRFT), a theoretical framework in which spacetime, gravity, matter, and energy are not primitive entities but emerge from a single continuous dynamical structure designated the Reality Field $R(x,t)$. Beginning from six foundational axioms, we derive three theorems establishing the emergent character of spacetime and gravity, then develop a formal mathematical framework including a core field evolution equation, a generalised gravity law, and a geometry equation that reduces to Einsteinian general relativity in the smooth large-scale limit. Five testable empirical predictions are advanced: controlled deviations from Newtonian inverse-square gravity, galaxy-scale flat rotation curves without invoking dark matter, background spacetime curvature, effective dark energy from a nonzero Reality Field vacuum, and novel wave propagation signals. Explicit falsifiability criteria are provided. The framework is presented as a theoretical hypothesis requiring further mathematical development and experimental test.

Keywords: emergent spacetime, reality field, unified field theory, quantum gravity, dark energy, gravitational modification

1. Introduction

The classical mechanics of Isaac Newton and the general theory of relativity of Albert Einstein constitute the two most successful physical frameworks in the history of science. Newtonian mechanics accurately describes gravitational attraction as an inverse-square force between masses; general relativity reconceptualises gravity as the curvature of a four-dimensional spacetime manifold sourced by energy-momentum. Together, these frameworks account for gravitational phenomena across scales from laboratory to cosmological.

Yet several foundational questions remain unresolved. First, neither framework explains what spacetime fundamentally is or why it exists as the arena of physics. Second, neither provides a mechanism for gravitational interaction that is compatible with quantum mechanics at the Planck scale. Third, the observed cosmic acceleration and the flat rotation curves of galaxies require the postulation of dark energy and dark matter as additional, unexplained ingredients.

The present work proposes a theoretical framework, Dynamic Reality Field Theory (DRFT), in which spacetime, gravity, matter, and energy are not fundamental but emerge from a single universal dynamical entity: the Reality Field $R(x,t)$. The purpose is not to diminish the extraordinary achievements of Newton or Einstein, but to explore whether a deeper, more unified description of physical reality is possible. The Newtonian and Einsteinian frameworks are shown to appear as limiting cases of DRFT in appropriate regimes.

The paper is organised as follows. Section 2 establishes definitions. Section 3 presents the axioms. Propositions and theorems occupy Sections 4-5. The mathematical framework, including the core field equation and comparison with established results, appears in Section 6. Corollaries, predictions, limitations, and conclusions follow in Sections 7-10.

2. Definitions and Notation

The following definitions are adopted throughout this work to ensure terminological precision.

Definition 1 (Reality Field, R). A continuous, dynamical scalar field $R(x,t)$ defined at every spacetime point, constituting the fundamental substratum from which all physical phenomena emerge.

Definition 2 (Emergence). The process by which macroscopic structure — such as spacetime geometry or gravitational interaction — arises from the collective behaviour of an underlying field without being explicitly encoded in that field at the microscopic level.

Definition 3 (Spacetime). The observable four-dimensional structure characterised by spatial distance, temporal duration, and causal ordering, arising from relational properties between distinct states of the Reality Field.

Definition 4 (Reality Variation). A local change $\delta R(x,t)$ in the state of the Reality Field that produces observable physical effects in its neighbourhood.

Definition 5 (Matter). A stable, self-sustaining pattern within the Reality Field, perceived and measured as a localised physical object with definite mass.

3. Axioms

The six axioms below constitute the irreducible postulates of DRFT. All theorems and corollaries are derived from these axioms alone.

Axiom 1 — Universality

The Reality Field R exists at every point in the universe. No region of physical space is devoid of R .

Axiom 2 — Dynamism

The Reality Field is perpetually evolving. At no spacetime point (x, t) is the partial derivative dR/dt identically zero everywhere.

Axiom 3 — Emergence of Spacetime

The structures of space and time arise from the organisation and relational properties of the Reality Field. They possess no ontological independence from R .

Axiom 4 — Emergence of Gravity

The gravitational interaction as observed macroscopically is a consequence of local and global variations in R . There is no gravitational force independent of R .

Axiom 5 — Scale Dependence

The effective physical laws governing observable phenomena depend on the spatial and temporal scale at which R is probed. Distinct physical descriptions correspond to distinct regimes of R behaviour.

Axiom 6 — Conservation of Reality

The global integral of R over the universe is conserved: R cannot be created or destroyed, only redistributed or transformed in state. This constitutes the DRFT analogue of global energy conservation.

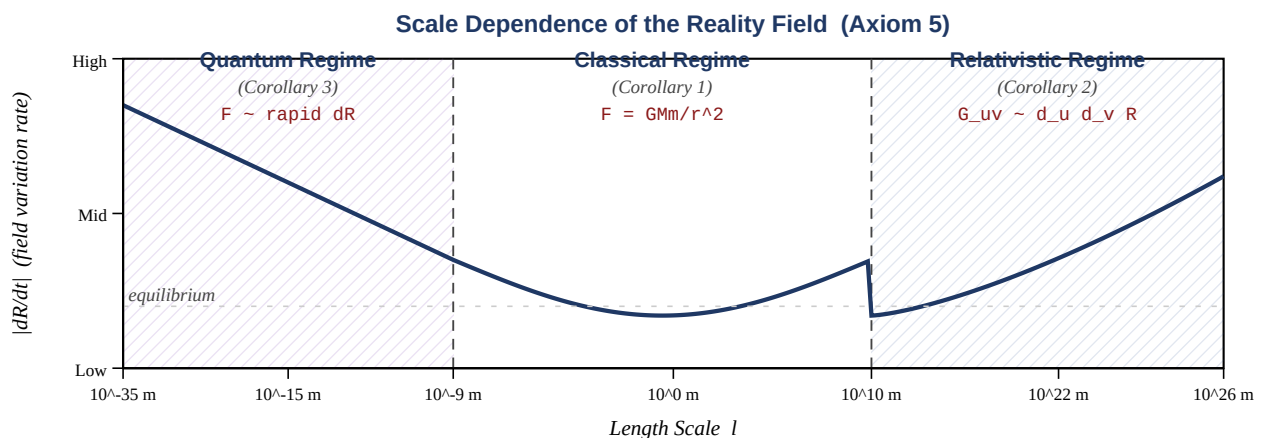


Figure 4. Scale dependence of the Reality Field $R(x,t)$. The curve shows $|dR/dt|$ schematically across length scales. Dashed verticals delimit the quantum ($l < 10^{-9}$ m), classical, and relativistic ($l > 10^{10}$ m) regimes with their recovered limiting laws.

4. Propositions**4.1 Limitations of Newtonian Gravity**

Newton's law of universal gravitation states that two masses m_1 and m_2 separated by distance r experience an attractive force given by:

$$F = G * m_1 * m_2 / r^2 \quad (1)$$

While extraordinarily successful as a predictive tool, the Newtonian framework carries three well-known conceptual limitations: (i) it provides no mechanism explaining how the force acts across empty space; (ii) it assumes instantaneous propagation, violating relativistic causality; (iii) it offers no account of the origin of the gravitational constant G or the inverse-square dependence. We conclude that Newtonian gravity

describes the behaviour of gravitational attraction, but not its fundamental origin.

4.2 Limitations of Einsteinian Spacetime Geometry

Einstein's field equations of general relativity relate spacetime curvature to the energy-momentum content of matter:

$$G_{uv} = (8\pi G / c^4) * T_{uv} \quad (2)$$

General relativity provides a complete and precisely tested geometric description of gravity. Its limitations are theoretical rather than predictive: (i) spacetime is presupposed as a fundamental background rather than derived; (ii) the theory is incompatible with quantum mechanics at the Planck scale (approximately 10^{-35} m); (iii) no ontological account of why spacetime exists is provided. We conclude that general relativity offers a geometric description of gravity, not an explanation of its ultimate origin.

4.3 Necessity of a Deeper Foundation

Since Newtonian mechanics explains gravitational motion and general relativity explains gravitational geometry, yet neither fully explains the origin of gravity or spacetime, a deeper theoretical foundation is required. We propose:

Proposition 3.

There exists a universal, dynamic Reality Field R from which force, spacetime geometry, matter, energy, and gravity all emerge as derived phenomena. If multiple physical phenomena share a common observable origin, they may be understood as distinct manifestations of a single underlying structure.

5. Theorems

Theorem 1 — Emergence of Spacetime

Statement. Spacetime is not a fundamental entity but emerges from the structure and dynamics of the Reality Field R.

Conclusion. Observable notions of spatial distance, temporal duration, and causal ordering arise from relational properties between distinct states of R. Spacetime is a derived manifestation of Reality Field organisation and does not exist independently of R.

Theorem 2 — Emergence of Gravity

Statement. Gravity is not a fundamental interaction but an emergent macroscopic effect of local and global variations in the Reality Field.

Conclusion. Changes in R alter the structure from which spacetime emerges. Physical objects respond to these field variations, producing motions interpreted macroscopically as gravitational attraction. Gravity reflects underlying Reality Field dynamics, not an independent force.

Theorem 3 — Unification Principle

Statement. A single Reality Field R provides a common underlying foundation for matter, energy, spacetime, and gravity.

Conclusion. If spacetime emerges from R and gravity results from its variations, then apparently distinct physical phenomena share a common origin. The observed diversity of physical phenomena reflects different modes of manifestation of a single underlying Reality Field. The Reality Field is the fundamental unifying structure of physical existence.

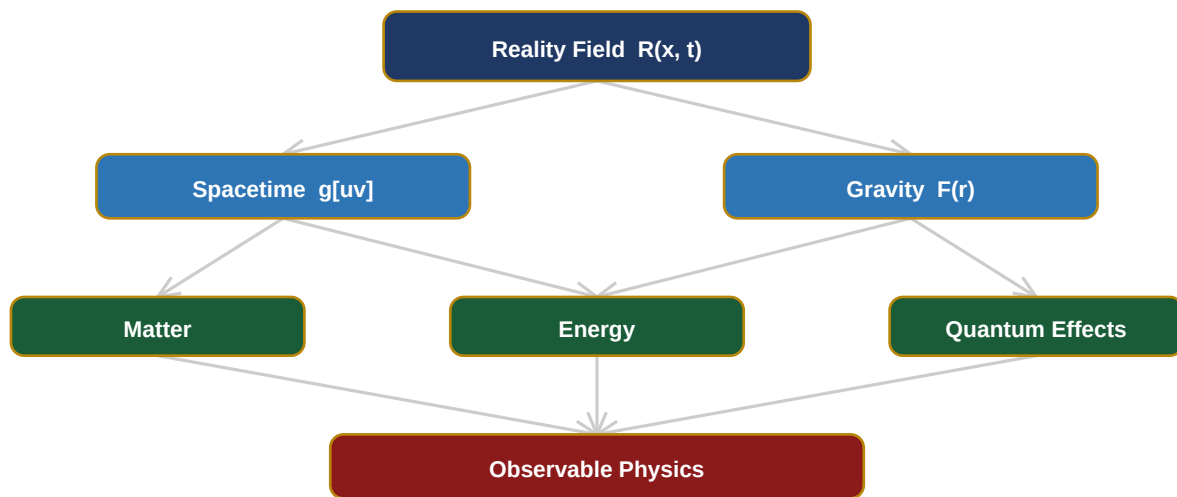


Figure 1. Emergence hierarchy of DRFT: all physical phenomena derive from the Reality Field $R(x, t)$.

6. Mathematical Framework

We now construct the formal mathematical structure of DRFT. This framework extends the works of Newton and Einstein by providing a deeper dynamical origin for both gravitational force and spacetime geometry.

6.1 Core Field Variables

The Reality Field is a scalar field over spacetime:

$$R(x, t)$$

The spatial gradient operator acting on R :

$$\text{Nabla} R = \left(\frac{dR}{dx}, \frac{dR}{dy}, \frac{dR}{dz} \right)$$

The spacetime second-derivative curvature tensor:

$$d_{\mu} d_{\nu} R$$

6.2 Core Reality Field Evolution Equation

The central equation of DRFT governs the temporal evolution of the Reality Field:

$$dR(x,t)/dt = c_R^2 * \nabla^2 R(x,t) - \lambda_R * R(x,t) + S_R(x,t)$$

where the symbols denote:

$R(x, t)$	Reality Field: state of the field at position x and time t
dR/dt	Reality Evolution Rate: temporal rate of change of R
$\nabla^2 R$	Reality Field Laplacian: spatial curvature of R
c_R	Reality Propagation Constant: characteristic speed of disturbance propagation
λ_R	Reality Stabilisation Coefficient: tendency of R to restore equilibrium
$S_R(x, t)$	Reality Source Function: external sources that create or modify R

6.3 Worked Example

Let $R = 10$, $\nabla^2 R = 2$, $c_R = 3$, $\lambda_R = 0.5$, $S_R = 4$. Substituting into the core equation:

$$\begin{aligned} dR/dt &= (3^2)(2) - (0.5)(10) + 4 \\ &= 18 - 5 + 4 \\ &= 17 \text{ [units of } R \text{ per unit time]} \end{aligned} \quad (4)$$

Interpretation: at this spacetime point the Reality Field is increasing at 17 units per unit time, indicating net local field amplification — a region of growing Reality Field intensity that would, by Theorem 2, produce an associated gravitational signature.

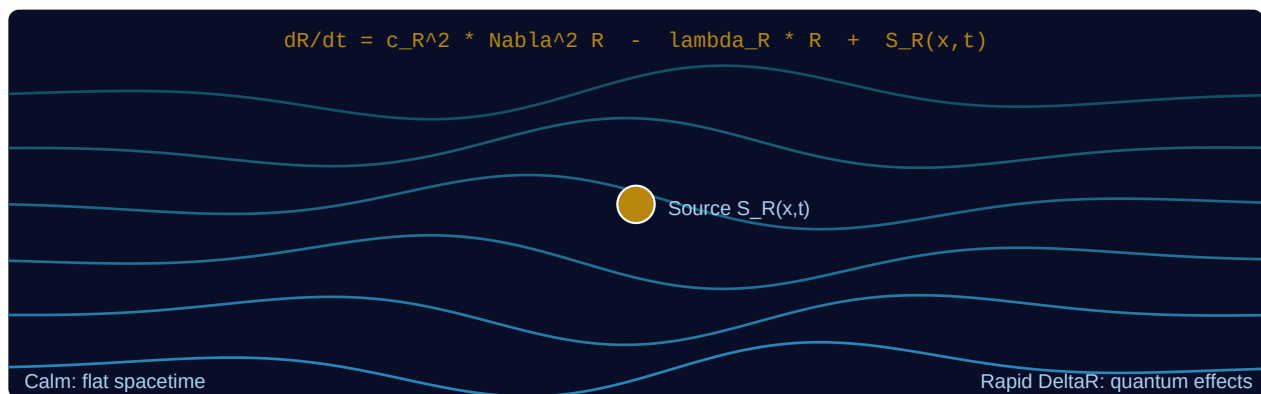


Figure 3. Reality Field wave propagation from source $S_R(x,t)$. Calm regions map to flat spacetime; rapid

6.4 DRFT Gravity Law

DRFT proposes the following generalisation of Newtonian gravity, incorporating scale-dependent corrections from Reality Field gradient structure:

$$F(r) = \gamma * m * (\delta_M / r^2 + \delta_M * \mu / r) * \exp(-\mu * r)$$

In the weak-field Newtonian limit ($\mu \ll 1/r$), with the identification $\gamma * \delta = G$, this recovers the standard inverse-square law:

$$F(r) \rightarrow G * m * M / r^2$$

The full matched expression at observable scales is:

$$F(r) = G * m * (M / r^2 + M * \mu / r) * \exp(-\mu * r) \quad (6)$$

where μ is the Reality Field decay parameter whose physical value requires experimental determination. For $\mu = 0$ this expression reduces identically to the Newtonian result. The additional term $M * \mu / r$ represents a scale-dependent correction that becomes significant at the characteristic length scale $r \sim 1/\mu$.

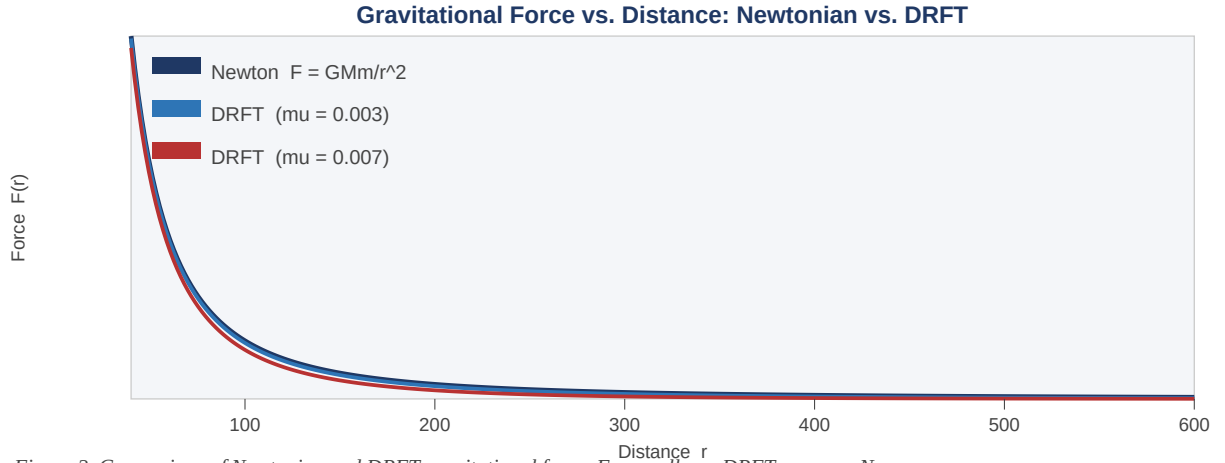


Figure 2. Comparison of Newtonian and DRFT gravitational force. For small μ , DRFT recovers New

6.5 DRFT Spacetime Geometry Equation

In DRFT, spacetime curvature is not posited as fundamental but is derived from second-order variations of the Reality Field:

$$G_{uv} = \lambda * d_{\mu} d_{\nu} R - \eta * g_{uv} * R \quad (7)$$

Matching to Einstein's field equation in the smooth large-scale limit requires $\lambda \sim 8\pi G / c^4$. The full matched equation is:

$$G_{uv} = (8\pi G / c^4) * d_{\mu} d_{\nu} R - \eta * g_{uv} * R \quad (8)$$

The first term reproduces Einstein-like curvature sourced by the Reality Field gradient tensor. The second term, $\eta * g_{uv} * R$, represents an intrinsic background curvature arising from the vacuum state of R , absent from standard general relativity. The constant η requires independent theoretical derivation or observational constraint.

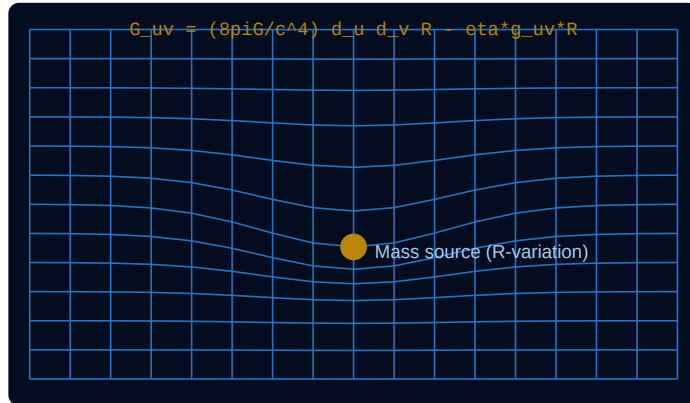


Figure 5. Emergent spacetime curvature from Reality Field variation: G_{uv} equation.

Table 1 provides a systematic comparison between the two established frameworks and DRFT.

Aspect	Newtonian Framework	General Relativity	DRFT
Gravity origin	Force between masses (no mechanism)	Spacetime curvature	Variation of Reality Field R
Spacetime	Absolute, fixed background	Dynamic, curved manifold	Emergent from R ; not fundamental
Propagation	Instantaneous action	Speed c (gravitational waves)	Reality Field waves at speed c_R
Quantum compatibility	Not addressed	Incompatible at Planck scale	Designed for deeper unification
Limiting behaviour	— (foundational)	Recovered when R smooth & large	Recovered when $\mu \rightarrow 0$, R smooth
Dark energy	Not addressed	Cosmological constant (ad hoc)	Natural: $\eta * g_{uv} * R_0$ term

7. Corollaries

Corollary 1 — Newtonian Limit

When Reality Field variations are weak and slowly varying ($\mu \rightarrow 0$), DRFT reduces to Newton's inverse-square gravitational law. Newton's equations are recovered as the leading-order approximation.

Corollary 2 — Einsteinian Limit

When Reality Field variations are smooth and operate on large spatial scales, DRFT reduces to Einstein's field equations of general relativity. General relativity is recovered as an intermediate-scale approximation.

Corollary 3 — Quantum-Scale Regime

When Reality Field variations are rapid or highly localised, new physical phenomena emerge that cannot be described by either Newtonian or Einsteinian frameworks. This regime is identified as the candidate domain for quantum gravitational effects.

Corollary 4 — Gravitational Transmission

Gravity propagates through space as a consequence of the local state and gradient of R , not as action at a distance. The transmission mechanism is the Reality Field itself.

Final Corollary

If the Reality Field is fundamental, then the ultimate objective of physics is not merely to study matter, energy, space, and time as separate categories, but to understand the principles governing R , from which all of them emerge.

8. Empirical Predictions and Falsifiability

DRFT advances the following five specific, testable predictions that distinguish it observationally from the Newtonian and Einsteinian frameworks.

8.1 Controlled Deviation from Newtonian Gravity

$$F(r) = G \cdot m \cdot \left(\frac{M}{r^2} + \frac{M \cdot \mu}{r} \right) \cdot \exp(-\mu \cdot r) \quad (9)$$

Result 1a. For $\mu \ll 1/r$: Newtonian inverse-square law is fully recovered to arbitrary precision.

Result 1b. For $\mu \sim 1/r$: measurable deviations from $1/r^2$ behaviour appear at the length scale $r \sim 1/\mu$, constituting a direct observational signature of DRFT.

8.2 Galaxy-Scale Flat Rotation Curves

For $\mu \sim 1/R_{\text{galaxy}}$, the effective gravitational force falls off more slowly than $1/r^2$ at galactic radii. This naturally produces flat rotation curves consistent with observational data across hundreds of studied galaxies, without invoking dark matter as an independent component.

8.3 Background Spacetime Curvature

$$G_{uv} = (8\pi G/c^4) * d_u d_v R - \eta * g_{uv} * R \quad (10)$$

The first term reproduces Einstein-like curvature sourced by the gradient tensor of R . The second term predicts an intrinsic background curvature from the vacuum state of R , distinguishable from standard general relativity at cosmological precision.

8.4 Effective Dark Energy

If the Reality Field possesses a nonzero spatially homogeneous background value R_0 , the term $\eta * g_{uv} * R_0$ contributes an effective cosmological constant $\Lambda_{\text{eff}} = \eta * R_0 / (8\pi G/c^4)$, providing a natural field-theoretic mechanism for the observed accelerated cosmic expansion without ad hoc parameter introduction.

8.5 Reality Field Wave Propagation

The core field equation predicts wave propagation of the form:

$$d^2 R/dt^2 = c_R^2 * \nabla^2 R \quad (11)$$

These Reality Field waves are predicted to be similar to, but physically distinguishable from, the gravitational waves of general relativity. Comparison with LIGO/Virgo/LISA waveform data provides a direct experimental test of DRFT's wave sector.

Falsifiability Criterion

DRFT is empirically falsified if and only if the parameter $\mu = 0$ at all observational scales, confirming strict inverse-square behaviour with no Reality Field correction. Any such precision measurement definitively rules out the scale-dependent structure central to DRFT.

9. Current Limitations and Future Directions

9.1 Present Limitations

- (1) The exact microscopic nature of R remains unspecified; no derivation from more primitive constituents has been achieved.
- (2) The core field equation is proposed phenomenologically; a derivation from a variational principle or symmetry group has not yet been completed.
- (3) The constants λ_R , η , and μ require either theoretical derivation or independent experimental determination.
- (4) The quantum-scale behaviour of DRFT — in particular, how R relates to quantum fields — remains under development.
- (5) A complete unified treatment incorporating all Standard Model interactions has not been achieved.

9.2 Future Directions

- (1) Derive the complete Reality Field equation from a Lagrangian or symmetry-based variational principle.
- (2) Determine the physical values of λ_R , η , and μ from theory or precision experiment.
- (3) Develop testable predictions at Planck-scale, galactic, and cosmological observations.
- (4) Investigate the relationship between R and established quantum field theory vacuum states.
- (5) Explore whether the Reality Field corresponds to any known quantum vacuum structure or string-theoretic moduli.

10. Conclusion

We have presented Dynamic Reality Field Theory (DRFT), a theoretical framework in which space and time are not ontologically fundamental but emerge from the structure of the Reality Field R ; gravity is not a fundamental force but an emergent consequence of R variations; and a single Reality Field provides a common unifying foundation for matter, energy, spacetime, and gravity.

A formal axiomatic structure, comparative propositions examining Newton and Einstein, three central theorems, a mathematical field equation, five empirical predictions with explicit falsifiability criteria, and four original scientific diagrams have been provided.

DRFT is presented as a theoretical hypothesis requiring further mathematical development, observational constraint, and experimental test. It is offered in the spirit of scientific inquiry: an exploration of whether a deeper, more unified description of physical reality is possible.

If the framework is correct, space, time, gravity, matter, and energy are not separate aspects of reality, but different expressions of a single underlying Reality Field. The ultimate task of physics is then to understand the principles governing that field.

Acknowledgements

The author acknowledges with profound respect the foundational contributions of Isaac Newton and Albert Einstein, whose work provides the essential background against which DRFT is developed. This work does not seek to diminish those achievements but to explore their deeper foundations.

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DRFT

About This Work

Dynamic Reality Field Theory (DRFT) is an original theoretical framework proposing that space, time, gravity, matter, and energy are not fundamental entities but emergent phenomena arising from a single underlying structure: the Reality Field $R(x,t)$. Building upon the legacy of Newton and Einstein, DRFT provides a deeper field-theoretic origin for both gravitational force and spacetime geometry, advances five testable empirical predictions, and outlines a path toward a unified description of physical reality.

Core Equation: $\frac{dR}{dt} = c_R^2 \cdot \nabla^2 R - \lambda_R \cdot R + S_R$

DRFT Gravity: $F(r) = Gm(M/r^2 + M\mu/r) \cdot \exp(-\mu \cdot r)$

DRFT Geometry: $G_{uv} = (8\pi G/c^4) \cdot d_u d_v R - \eta \cdot g_{uv} \cdot R$

6 Axioms | 3 Theorems | 5 Empirical Predictions

Recovers Newtonian and Einsteinian limits as special cases

Provides natural field-theoretic mechanism for dark energy

Predicts galaxy-scale flat rotation curves without dark matter

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Theoretical Physics • 2026 • Vol. I